A short summary for community feedback

Matching and learning for recognising isolated signs from hand landmarks

[Researcher name, affiliation and contact: to be completed before sending]

Why we are writing. I am a hearing researcher, and this project was not led by Deaf people. Before I submit a short methods paper, I would like your critique of how I describe what the work is and is not, and whether anything in it could mislead or cause harm. I would like to pay for your time. You are free to say that the work should not be published in this form.

What the study is. It compares two ways for a computer to label a short video of one ASL sign from a fixed list: (1) comparing the sign with a few stored examples ("template matching"), and (2) small neural networks trained on examples. Both work from the positions of hands and body that a standard tool (MediaPipe) finds in each video frame; they do not look at faces. I used two published datasets with consent from Deaf signers (PopSign and ASL Citizen) and only isolated signs from vocabularies of up to 300 signs.

What I found. The trained networks were more accurate than template matching at every number of examples I tried, including one example per sign. For example, with 50 signs and 10 examples per sign the best network was right about 61.1% of the time against 38.6% for template matching on new signers. With one example per sign and 50 signs no method exceeded 23.7%. These are results on two datasets, for ASL only.

What the study is not. It is not sign language translation or interpretation. It does not understand grammar, facial expressions or context. It says nothing about whether any such system helps Deaf people, and it does not claim to. Accuracy on isolated signs is not communication access. I do not want this work used as a reason to replace interpreters or other accommodations.

Data and consent. The videos belong to the dataset owners; I deleted them after extracting hand and body positions, kept no face information from ASL Citizen, will not share ASL Citizen derived data, and make no attempt to identify anyone. I do not know how the participants feel about this specific use beyond the consent they gave the dataset creators.

What I would like to ask you.

1. Is it clear what this work is and is not? Is any sentence in the abstract or discussion misleading, or likely to be read as a claim of benefit?
2. Is there Deaf-led research, a community position or a better framing I have missed?
3. Is there any use of results like these that worries you, and should I say so in the paper?
4. Should this be published, and if so, anything you would want changed or added?

What I will do with your answers. I will record who was asked, what changed because of the feedback and what did not change and why. I will name you in the paper only if you agree. If I

cannot get feedback before submission, the paper will say so instead of implying that the community was consulted.

Contact and payment. [contact details; offer of payment for reviewers' time to be completed by the researcher]